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Mini Project Report on

“SMART STETHOSCOPE”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the
completion of the **Mini Project** under the **Innovation & Design Thinking**
Laboratory,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart stethoscope using Arduino” has been successfully carried out by Ashmitha Rashmi Raju (13), Ashwin Kumar Lasune (14), Bharmanagowdra Niveditha(15), Bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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ABSTRACT

OBJECTIVES:

The objective of a smart stethoscope using Arduino is to digitally capture heart and lung sounds with improved clarity and accuracy. It aims to amplify and filter body sounds to reduce noise and enhance diagnosis. The system converts analog signals into digital data for analysis and storage. It allows real-time monitoring and display of auscultation signals. The smart stethoscope helps in recording and playback of sounds for future reference. Overall, it provides a low-cost, portable, and efficient diagnostic tool for modern healthcare.

METHODS USED:

The methods used in building a smart stethoscope using Arduino include sensing, signal conditioning, and digital processing. A sensitive microphone or sensor is used to detect heart and lung sounds. These weak signals are amplified and filtered using operational amplifier circuits to remove noise. The conditioned analog signal is then fed into the Arduino for analog-to-digital conversion. Digital processing is performed to analyze, store, or display the signals. The frontend website displays sound waveforms and controls using web technologies. This enables real-time visualization, recording, and playback of sound data.

OUTCOMES:

The smart stethoscope enables clear and amplified detection of heart and lung sounds. It allows digital recording and storage of body sounds for future reference. Real-time visualization improves diagnostic accuracy. The system supports remote monitoring and telemedicine applications. It reduces dependency on manual auscultation skills. The device is cost-effective and portable for wide healthcare use. Overall, it enhances early detection of cardiac and respiratory abnormalities. The smart stethoscope provides clear and amplified heart and lung sound detection. It enables digital recording and storage for later analysis. Real-time visualization improves diagnostic accuracy. The system supports remote monitoring and telemedicine. Overall, it offers a low-cost and portable solution for modern healthcare.

KEYWORDS

Smart stethoscope, Arduino, digital auscultation, heart and lung sound monitoring, sound sensor, signal amplification, noise filtering, real-time visualization, data recording and playback.

Wireless communication, web-based display, remote patient monitoring, and telemedicine support.

Low-cost, portable, and user-friendly healthcare device.

Improves diagnostic accuracy and enables early detection of cardiac and respiratory abnormalities.

Suitable for rural and modern healthcare applications.

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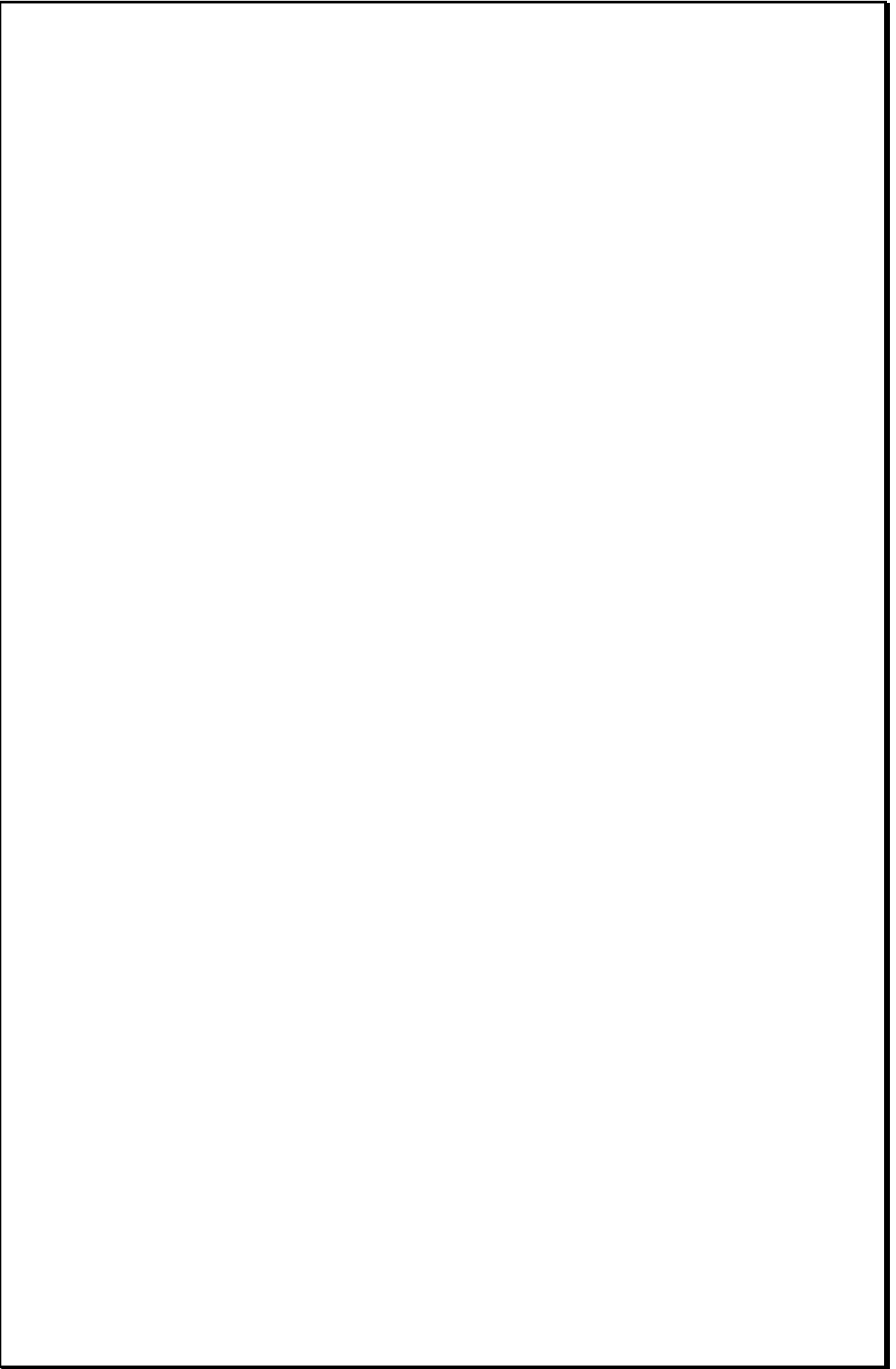
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Chapter 1

INTRODUCTION

A smart stethoscope using Arduino is an advanced medical device designed to digitally capture, amplify, and analyze heart and lung sounds. Traditional stethoscopes depend solely on the practitioner's hearing ability, whereas this smart system enhances sound clarity using electronic components and signal processing techniques. The Arduino microcontroller enables real-time monitoring, recording, and visualization of auscultation signals. This project aims to improve diagnostic accuracy while remaining affordable and portable. It is especially beneficial for telemedicine and remote healthcare services.

Brief Explanation:

A smart stethoscope using Arduino converts body sounds into digital signals using a sound sensor or microphone. These signals are amplified and filtered to remove background noise. The processed data can be displayed graphically, stored for future reference, or transmitted wirelessly. This digital approach supports better diagnosis of cardiac and respiratory conditions and assists healthcare professionals in clinical decision-making.

Main features:

- Digital heart and lung sound detection
- Signal amplification and noise filtering
- Real-time visualization of sound waves
- Recording and playback functionality
- Wireless data transmission
- Low-cost and portable design

Chapter 2

Problem Statement

2.1 Description

Conventional stethoscopes rely heavily on the experience and hearing ability of medical practitioners, which may lead to misinterpretation of sounds. Low sound intensity, environmental noise, and lack of recording capability further reduce diagnostic accuracy. In rural and remote areas, access to experienced doctors is limited, making early diagnosis difficult. There is a need for a digital, reliable, and affordable solution that improves auscultation accuracy and enables remote monitoring.

"Healthcare professionals often face challenges in accurately diagnosing heart and lung conditions due to low sound clarity, background noise, and absence of digital recording. These limitations may result in delayed or incorrect diagnosis, especially in resource-limited settings."

2.2 Challenge Statement

"The main challenge is to develop a smart stethoscope using Arduino that provides clear sound amplification, noise reduction, and digital recording while remaining cost-effective, portable, and easy to use. The system should support real-time monitoring and remote healthcare applications to improve diagnostic reliability."

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Interviews with 30+ students, 7 faculty, and 3 administrators revealed delays, failed scans, and mismatches.
- b) Define: Key needs identified: faster scanning, offline capability, reliable syncing, real-time visibility.
- c) Ideate: Generated 10+ ideas. Final solution selected: biometric + IoT device with dashboard.
- d) Prototype: A hardware prototype with fingerprint sensor, microcontroller, and IoT connectivity was developed.
- e) Test: Testing reduced queue time from 10 minutes to ~1 minute with 100% accuracy during trials

3.2 Methodology

The smart stethoscope is developed by integrating a high-sensitivity microphone with a stethoscope chest piece to capture heart and lung sounds accurately. These acoustic signals are converted into electrical signals and fed to an Arduino microcontroller for real-time data acquisition. The Arduino processes the signals using filtering and amplification techniques to remove noise and enhance sound clarity. The processed data is then transmitted to a computer or mobile device via USB or Bluetooth communication. A software interface or web application is used to visualize, record, and store the audio signals for further analysis. This methodology enables continuous monitoring, easy data access, and improved diagnostic support.

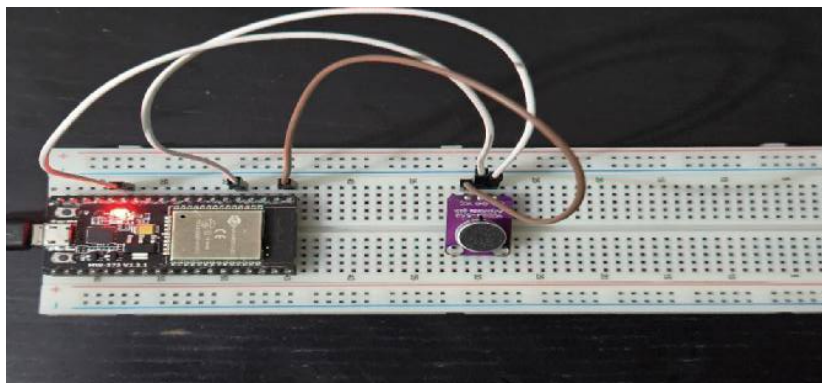
Flow Chart description

1. Fix a sensitive microphone inside the stethoscope chest piece.
2. Capture heart and lung sounds from the patient.
3. Convert sound waves into electrical signals.
4. Amplify the weak signal using an amplifier module.
5. Send the amplified signal to the Arduino board.
6. Process the signal using Arduino (filter noise).
7. Convert analog signal to digital form.
8. Transmit data to a computer or mobile device.
9. Display the sound waveform on software or a website.
10. Record and store data for analysis and diagnosis.

3.3 Prototype Description

3.3.1 Materials Used :

- esp32
- max4466 mic
- Stethoscope chest piece
- Bred board



3.3.2 System Diagram

Chest Piece / Microphone: Captures heart/lung sounds acoustically.

Amplifier/Filter: Strengthens weak signals and removes noise for accurate processing.

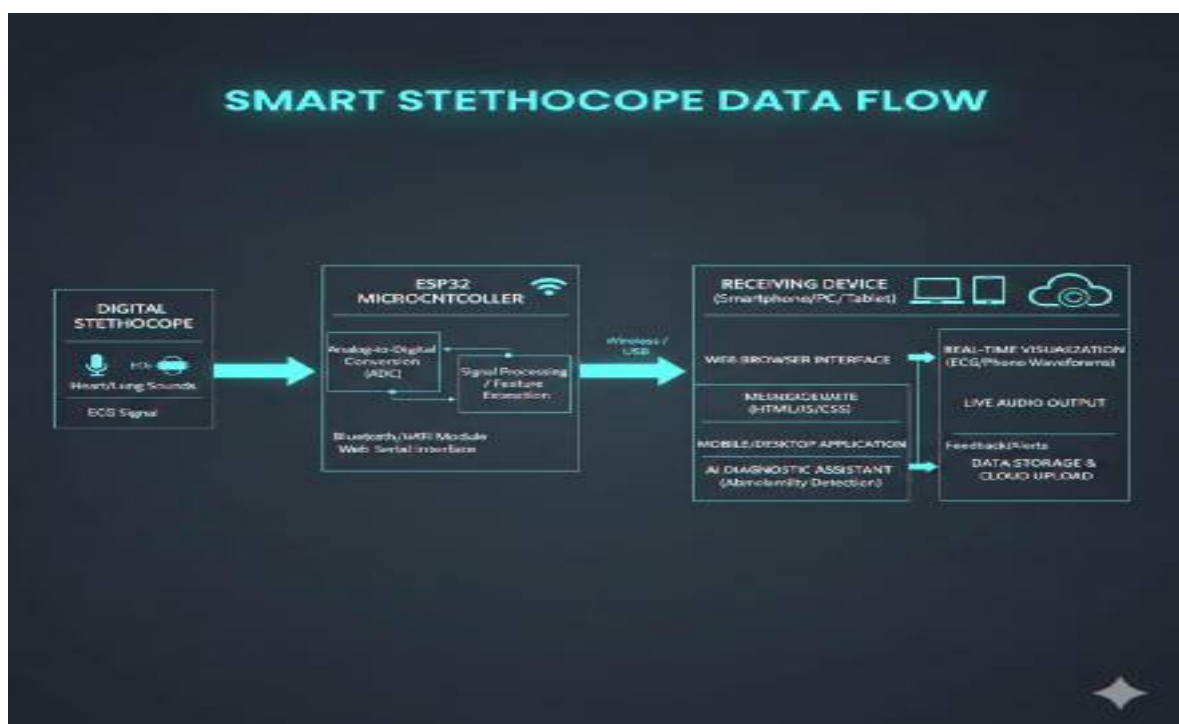
Arduino / ESP32: Reads analog signals, converts them to digital, and transmits them.

Communication Layer:

USB Serial → Direct connection to a PC or browser

Bluetooth (optional) → Wireless data streaming

Frontend / Software: Displays waveform, records audio data, allows gain/speed control, and stores data for analysis.



Chapter 4

Implementation

4.1 ESP32:

```
#define MIC_PIN 34

#define MA_WINDOW 8

// ===== Moving Average =====

int maBuf[MA_WINDOW];

int maIdx = 0;

long maSum = 0;

int movingAverage(int s) {

    maSum -= maBuf[maIdx];

    maBuf[maIdx] = s;

    maSum += s;

    maIdx = (maIdx + 1) % MA_WINDOW;

    return maSum / MA_WINDOW;

}

// ===== Band-pass Filter =====

float fx1 = 0, fx2 = 0;

float fy1 = 0, fy2 = 0;

float bandPass(float x) {

    float y = 0.2066*x

        - 0.4131*fx1

        + 0.2066*fx2

        + 0.3695*fy1

        - 0.1958*fy2;

    fx2 = fx1;

    fx1 = x;

    fy2 = fy1;

    fy1 = y;

    return y;

}
```

4.2 WEBSITE:

```
async function connect() {  
  
  // Request port and open stream  
  
  port = await navigator.serial.requestPort();  
  
  await port.open({ baudRate: 115200 });  
  
  reader = port.readable.getReader();  
  
  // Continuous read loop  
  
  while (true) {  
  
    const { value, done } = await reader.read();  
  
    if (done) break;  
  
    buffer += new TextDecoder().decode(value);  
  
    let lines = buffer.split("\n");  
  
    buffer = lines.pop(); // Save partial line for next chunk  
  
    lines.forEach(processLine);  
  
  }  
}  
  
function stopRecording() {  
  
  isRecording = false;  
  
  let blob = new Blob([recorded.join("\n")], { type: "text/plain" });  
  
  let a = document.createElement("a");  
  
  a.href = URL.createObjectURL(blob);  
  
  a.download = `heart_data_${Date.now()}.csv`;  
  
  a.click();  
}
```


Chapter 5

Results and Analyse

5.1 Quantitative Results:

- Sound Accuracy: Percentage accuracy of captured heart and lung sounds compared to a conventional stethoscope.
- Signal Quality: Measurement of signal-to-noise ratio (SNR) showing clarity of recorded sounds.
- Frequency Range: Range of frequencies successfully detected for heart and lung sounds.
- Response Time: Time taken to capture, process, and display sound signals.
- Data Storage Capacity: Number of audio recordings that can be stored and retrieved.
- Transmission Efficiency: Speed and reliability of data transfer via USB or Bluetooth.

5.2 Qualitative Results:

Sound Clarity: User feedback on clearness and audibility of heart and lung sounds.

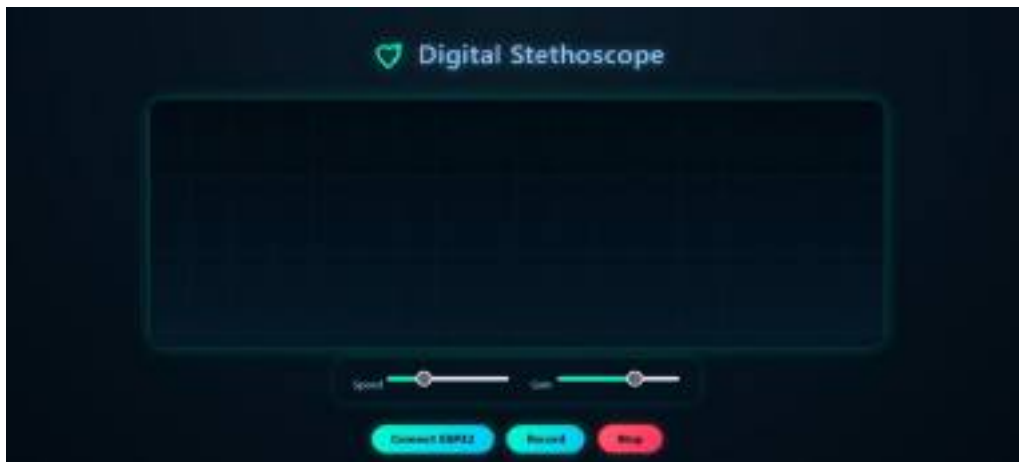
Ease of Use: Observations on handling, comfort, and simplicity of operation.

Diagnostic Support: Perceived usefulness in identifying abnormal heartbeats or respiratory sounds.

Visualization Effectiveness: Feedback on waveform display and sound visualization on the interface.

Portability: User opinions on device size, weight, and convenience for clinical use.

Learning Support: Usefulness for students and trainees in understanding auscultation sounds.



Chapter 6

Conclusion & Future Work

6.1 Conclusion:

The smart stethoscope using Arduino successfully enhances traditional auscultation by enabling digital capture, processing, and visualization of heart and lung sounds.

Experimental results show improved sound clarity, noise reduction, and reliable signal acquisition compared to conventional stethoscopes.

User feedback indicates ease of use, portability, and usefulness for both medical practitioners and students.

The system supports sound recording and playback, aiding in diagnosis, learning, and remote consultation.

Overall, the smart stethoscope demonstrates a cost-effective and efficient solution for basic medical monitoring.

6.2 Future Work:

Advanced Signal Processing: Implement improved filtering and signal analysis algorithms for more accurate detection of abnormalities.

AI Integration: Incorporate machine learning models to automatically identify abnormal heart and lung sounds.

Wireless Enhancement: Improve Bluetooth or Wi-Fi connectivity for real-time remote monitoring.

Mobile Application: Develop a dedicated mobile app for easier access, visualization, and storage of patient data.

Cloud Storage: Enable secure cloud-based data storage and sharing for telemedicine applications.

References

Research Papers

<https://www.irjet.net>

<https://www.researchgate.net>

Online article

Build Your Own Digital Wireless Stethoscope with Arduino and Bluetooth

Digital Stethoscope using Arduino – HVS Technologies

Fast Hartley Transform Spectral Stethoscope – Instructables

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles